

MONARCH PASS, CO, USA

WMO: 722203

Lat: 38.497N

Lon: 106.319W

Elev: 12031

StdP: 9.33

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 03041

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-7.5	-3.6	-17.4	3.4	8.0	-13.2	4.2	9.1	61.5	13.7	56.6	12.0	27.6	250	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.9	64.5	43.6	63.4	43.0	61.3	42.7	48.5	56.9	47.3	55.8	46.0	54.7	16.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
45.4	71.1	49.8	44.7	69.0	49.3	43.0	64.6	48.4	24.0	56.9	23.1	55.7	22.2	54.0	53.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 52.4	47.7	43.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	32.0	15.0	14.4	21.6	27.4	36.1	48.6	53.1	50.4	44.9	32.9	24.1	14.8		
	DBStd	15.84	8.55	7.83	8.62	7.75	8.06	6.40	3.87	4.12	6.04	8.71	8.75	9.30		
	HDD50	6802	1083	998	880	678	433	98	16	46	167	532	778	1093		
	HDD65	12032	1548	1418	1345	1128	896	492	367	452	604	996	1228	1558		
	CDD50	243	0	0	0	0	1	56	113	59	13	1	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	22.1	25.4	28.8	25.8	24.9	20.1	18.8	14.0	13.5	17.8	23.2	25.9	27.7		
	PrecAvg	11.7	0.6	0.6	0.9	1.1	1.0	0.7	1.6	1.9	1.2	0.9	0.6	0.6		
	PrecMax	14.6	1.6	1.7	1.8	2.4	3.0	1.6	2.8	3.0	2.3	2.3	1.3	1.8		
(15) Precipitation	PrecMin	6.5	0.1	0.2	0.2	0.3	0.1	0.1	0.6	0.6	0.2	0.2	0.0	0.2		
	PrecStd	1.8	0.4	0.4	0.5	0.7	0.7	0.4	0.7	0.7	0.5	0.5	0.3	0.4		
	DB	36.1	34.1	44.6	48.2	58.9	68.2	69.8	66.4	63.5	55.1	46.1	35.9			
	MCWB	24.9	26.3	30.4	32.1	37.7	43.8	44.6	44.3	42.9	41.8	33.7	26.1			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	31.8	30.5	39.0	45.1	54.5	64.1	65.9	63.4	60.6	51.7	40.9	32.1			
	2% MCWB	24.0	23.9	27.6	30.6	36.2	41.9	44.5	43.4	41.8	39.4	30.2	24.6			
	5% DB	28.4	27.8	36.0	41.3	51.8	62.9	63.6	61.2	56.8	46.3	36.9	29.6			
	5% MCWB	22.0	22.3	26.1	29.3	35.3	41.1	44.1	43.1	40.2	35.4	27.8	23.6			
	10% DB	27.1	25.3	32.4	37.2	47.6	60.5	61.4	58.8	54.3	44.6	34.4	27.2			
	10% MCWB	21.2	19.8	24.6	27.6	33.9	40.3	44.3	43.3	39.6	33.3	26.9	21.4			
	0.4% WB	28.2	28.2	31.0	34.3	39.5	47.6	50.8	49.6	46.5	47.2	36.6	29.5			
	0.4% MCDB	31.0	31.0	41.5	43.8	52.9	58.0	58.7	58.2	55.7	50.7	43.8	33.3			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	25.9	25.8	28.9	32.3	37.8	45.1	49.1	47.9	44.5	41.0	31.4	26.2			
	2% MCDB	29.6	29.5	37.0	40.1	50.3	58.6	57.5	55.8	54.0	47.1	37.5	29.7			
	5% WB	23.4	22.9	27.2	30.7	36.3	43.6	48.0	46.6	43.3	37.3	29.6	24.1			
	5% MCDB	28.2	27.0	33.7	38.0	47.3	56.6	56.7	54.3	52.6	44.5	35.3	28.2			
	10% WB	21.6	21.1	25.5	29.2	34.9	42.0	46.8	45.2	41.9	34.9	27.8	22.1			
	10% MCDB	25.3	24.3	31.4	36.2	44.8	54.7	55.6	53.1	50.2	40.6	33.3	25.6			
	MDBR	10.6	10.9	12.7	13.7	14.8	18.3	16.9	16.4	15.9	13.6	11.2	10.4			
	5% DB	12.0	12.9	14.4	15.7	18.8	19.6	18.6	18.1	18.2	15.8	12.7	11.6			
(36) Mean Daily Temperature Range	MCWBR	8.5	8.8	8.2	8.6	8.6	8.9	7.3	7.6	7.7	8.0	7.1	7.5			
	MCDBR	10.9	11.9	13.4	14.0	16.7	18.8	16.5	15.7	15.3	14.2	11.8	10.2			
	MCWBR	8.6	8.6	8.2	8.3	8.2	8.7	8.1	7.8	7.0	7.5	7.5	7.7			
	MCDBR	8.6	8.6	8.2	8.3	8.2	8.7	8.1	7.8	7.0	7.5	7.5	7.7			
(40) Clear-Sky Solar Irradiance	taub	0.188	0.193	0.212	0.229	0.241	0.244	0.276	0.266	0.236	0.217	0.198	0.189			
	taud	2.606	2.598	2.630	2.647	2.682	2.694	2.617	2.632	2.766	2.773	2.728	2.630			
	Ebn at Noon	333	342	340	336	330	327	316	318	327	328	327	326			
	Edh at Noon	21	25	27	29	28	28	30	29	23	21	19	19			
(44) All-Sky Solar Radiation	RadAvg	873	1164	1605	1909	2081	2329	2025	1840	1682	1305	959	763			
	RadStd	48	67	120	111	156	141	113	86	101	90	57	48			

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+112	+29

Nomenclature: See separate page